

Reef Stress Project Report

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Heat stress is the main driver of coral bleaching, and when a reef is surveyed matters almost as much as how hot it got. The final model predicts which reefs bleach with an accuracy (AUC) of 0.76-0.89 on data it never saw, up from 0.68-0.81 for the first model. It rates 85% of 2,649 reef sites at high or very high bleaching risk in 2024, matching the fourth global bleaching event, and 73-90% in each year from 2027 to 2031. Among other threats, only crown-of-thorns starfish show a measurable link to lower coral cover.

Goal and data

The project set out to answer two questions: which threats are linked to reef damage, and which reefs are most likely to bleach in coming years. It combines four public data sources into one table of reef surveys, then builds prediction files for an interactive map.

Source	What it gives	Coverage used
Global Coral Bleaching Database (van Woesik et al.)	Diver surveys: % of coral bleached, coral cover, depth	5,294 surveys, 2013-2020
NOAA Coral Reef Watch	Satellite heat stress (degree heating weeks, DHW) and sea temperature at each reef's 5 km pixel	Weekly 2012-2026; monthly 1985-2026
Global Fishing Watch	Industrial fishing hours near each reef	Monthly 0.1° grid, 2012-2020
GBIF	Sightings of lionfish and crown-of-thorns starfish	All records to 2026

Degree heating weeks (DHW) measure accumulated heat above a reef's normal summer maximum. NOAA treats 4 DHW as the point where bleaching becomes likely and 8 DHW as severe. The target throughout is "significant bleaching": 10% or more of a reef's coral bleached.

The original plan used GCRMN survey data, but it is not publicly available, so the project switched to the Global Coral Bleaching Database. Its surveys stop in August 2020, so anything after 2020 is a prediction that cannot yet be checked against divers' observations.

How the project progressed

The project went through 13 stages, from a first model that failed on new years to a tested model and a five-year forecast. Stages are listed oldest first.

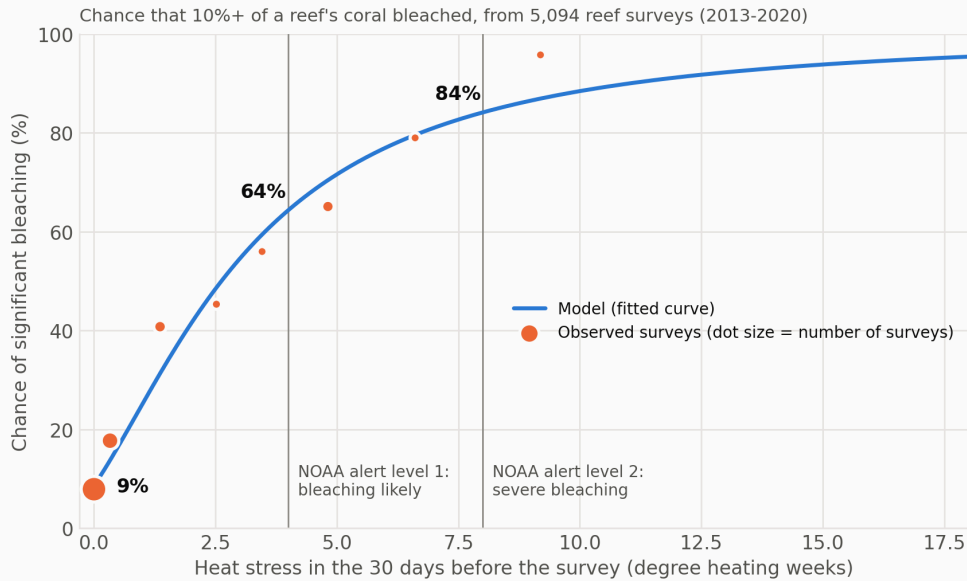
#	Stage	What was done	Result
1	Data pipeline	Joined surveys with NOAA heat stress and fishing effort for the 12 months before each survey	5,294 surveys; a merge bug that inflated rows to 6,670 was found and fixed
2	First model	Random forest predicting % of coral bleached	Failed on held-out years (R^2 below 0): worse than guessing the average
3	Heat model (v1)	Found bleaching tracks heat in the 30 days before the survey; switched to a logistic model on that alone	Accuracy 0.76 on held-out years; 2021-2026 predictions and a 2026-2030 scenario
4	Fishing	Added Global Fishing Watch hours within 10, 25 and 50 km	No detectable link to coral cover ($p = 0.175$)
5	Invasive species	Added GBIF lionfish and crown-of-thorns sightings	Crown-of-thorns linked to lower coral cover ($p = 0.011$); lionfish not significant ($p = 0.080$)
6	First charts	Six slide-ready charts	Later replaced (stage 13)
7	XGBoost	Gradient boosting on six temperature measures	No better than v1 (best 0.765 vs 0.762); the six-measure version did worse (0.710)
8	Forward-in-time tests	Trained on early years, tested on later years	v1 held up (0.68-0.70) but overpredicted bleaching 2.3-3.1×

#	Stage	What was done	Result
9	Depth	Added survey depth	Worse in all three tests; dropped
10	New model (v2)	Added heat timing, ocean-specific heat thresholds, and trained on bleaching severity	Better on every test (0.76-0.89); overprediction cut to 1.4-1.9×
11	Finer regions, blends	5° and 10° regional thresholds, boosted trees, model blends	Better on unseen reefs but worse on future years; rejected
12	Heat forecast	Rebuilt heat history back to 1985 from monthly sea temperature; forecast peak heat 1-5 years ahead with El Niño	Ranked reefs better than assuming the last five years repeat (0.756 vs 0.671)
13	Bleaching outlook	Fed the heat forecast through v2; moved threats and charts to v2	2027-2031 risk for every reef; seven updated charts

What drives bleaching

The chance of significant bleaching rises from 9% with no heat stress to 64% at 4 DHW and 84% at 8 DHW. The heat that matters most is in the 30 days before a survey, not the whole year.

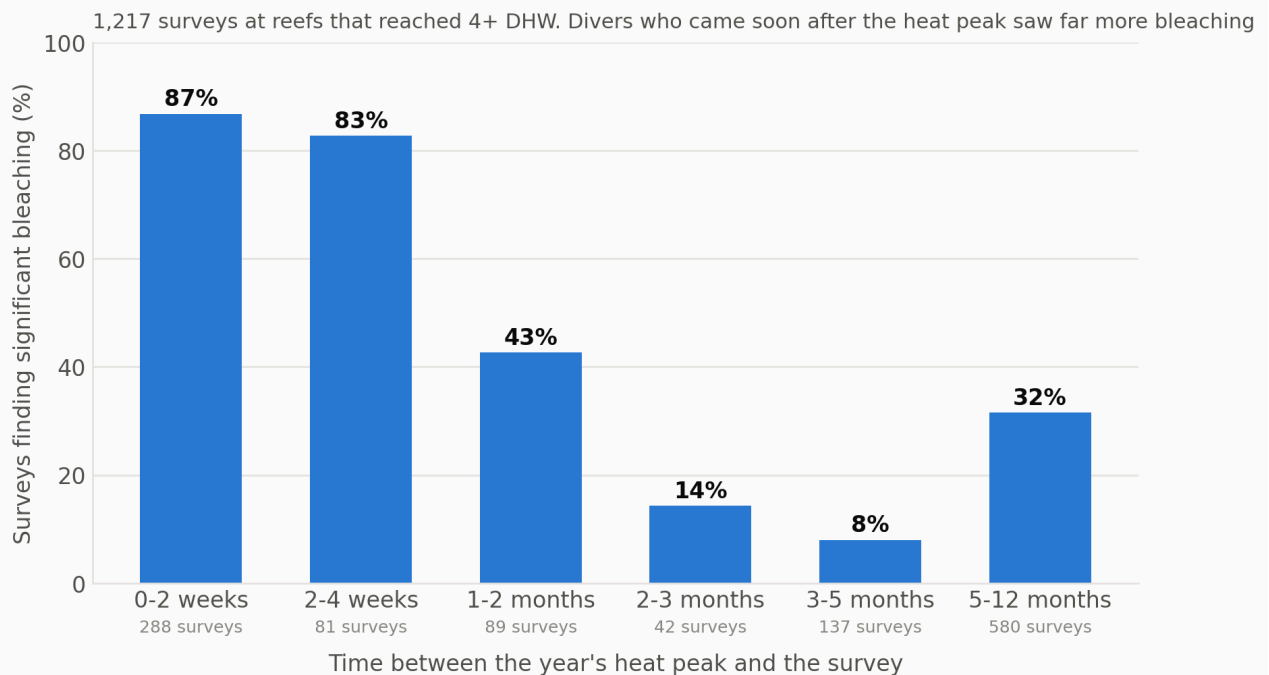
More heat stress, more bleaching



Data: Global Coral Bleaching Database (van Woesik et al.); NOAA Coral Reef Watch 5 km DHW. Curve: heat alone (logistic on $\log(1 + \text{DHW})$); the full model also uses timing and ocean.

Timing turned out to be the biggest missing piece. At reefs that reached 4+ DHW, surveys within two weeks of the heat peak found significant bleaching 87% of the time. Surveys two to three months later found it only 14% of the time, and three to five months later 8%.

Timing matters: bleaching fades within weeks

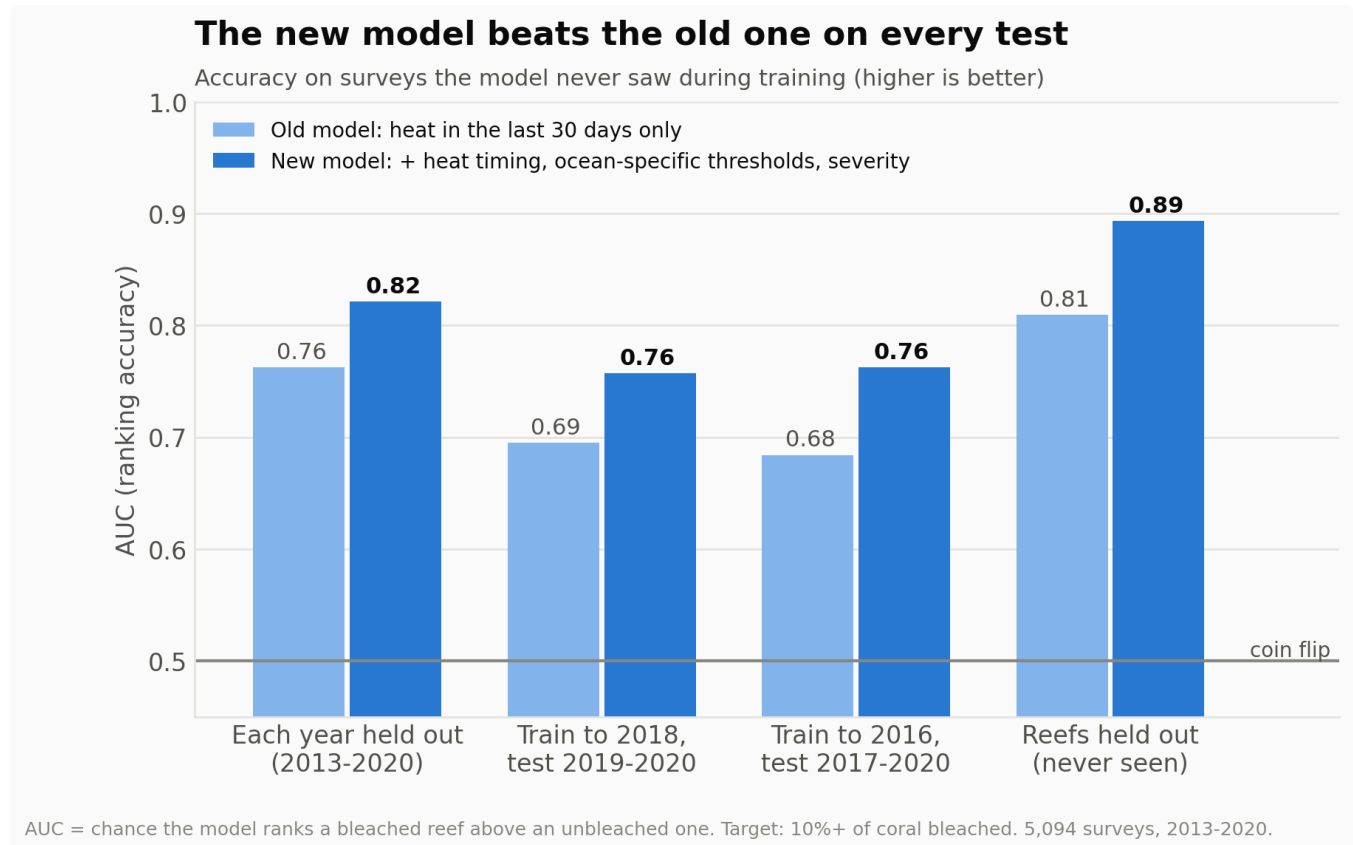


Bleaching peaks a few weeks after the heat peak, then corals recover or die. By 5-12 months the next warm season is often building again. The new model uses survey timing; the old one did not, so it overpredicted.

Bleaching is visible for only a few weeks: corals then either recover their color or die. The rise to 32% at 5-12 months reflects the next warm season often building again. The first model ignored timing, which is the main reason it overpredicted bleaching.

The bleaching model and how it was tested

The final model (v2) beats the first model (v1) on all four tests, by 0.06-0.08 in AUC. AUC is the chance the model ranks a bleached reef above an unbleached one: 0.5 is a coin flip, 1.0 is perfect.



Every model was scored only on surveys it never saw during training:

- **Each year held out:** train on seven years, test on the eighth, repeat for every year 2013-2020.
- **Train to 2018, test 2019-2020** and **train to 2016, test 2017-2020:** the model predicts a future it has not seen, which is how it is used.
- **Reefs held out:** test on reef sites the model never saw.

v2 uses three things v1 did not:

1. **Heat timing:** heat on the survey day, whether heat was still rising, days since the heat peak, and recent sea-temperature anomaly.
2. **Ocean-specific thresholds:** each ocean gets its own heat response, because Red Sea, Caribbean and Pacific reefs react differently to the same heat.
3. **Bleaching severity:** it learns from the full % bleached (a ridge regression), not only yes or no, then converts that to a chance of 10%+ bleaching.

v1 predicted about 22-24% of later-year surveys would show significant bleaching when only 7-10% did (2.3-3.1× too high). v2 cuts that to 13-15% predicted (1.4-1.9×). Rescaling the level to recent years was tried but was unstable, so v2's percentages are still somewhat high; its ranking of reefs is the reliable part.

Tried and rejected:

Approach	Result	Decision
Random forest on % bleached	Worse than guessing the average on new years	Dropped
XGBoost on six temperature measures	0.710 vs 0.762 for v1 (0.33 in 2018)	Dropped
Survey depth	Worse in all three time-based tests	Dropped
5° or 10° regional thresholds	Up to 0.916 on unseen reefs, but 0.715-0.728 on future years	Dropped: learns place-and-year patterns that don't carry forward
Blend of v2 and boosted trees	0.817 / 0.746 / 0.765 vs v2's 0.821 / 0.757 / 0.762	Dropped: no consistent gain

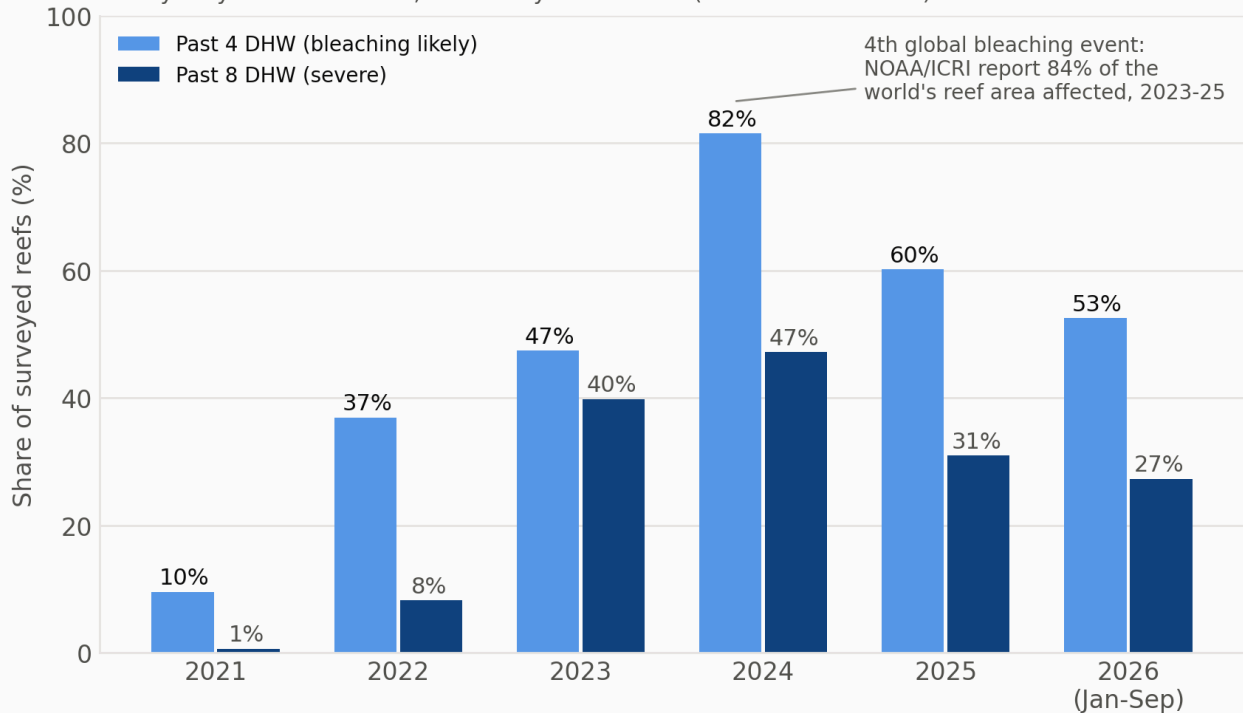
The 2017-2020 test years were reused many times while choosing the model, so the scores are probably slightly optimistic. After v2 the model was frozen.

Heat and bleaching risk, 2021-2026

2024 was the worst year: 82% of reef sites passed 4 DHW and 47% passed 8 DHW, and the model rates 85% at high or very high bleaching risk. That matches NOAA Coral Reef Watch's finding that 84% of the world's reefs had bleaching-level heat stress between January 2023 and March 2025, the fourth global bleaching event ([ICRI](#)).

Heat stress at reefs, 2021-2026

Peak yearly heat stress at 2,649 surveyed reef sites (NOAA satellite data)



Estimated from measured satellite heat stress, not diver surveys. 2026 covers January to September only.

These figures come from measured satellite heat, not from diver surveys, which end in 2020. Each reef-year is predicted for a survey two weeks after that year's heat peak, when bleaching is most visible.

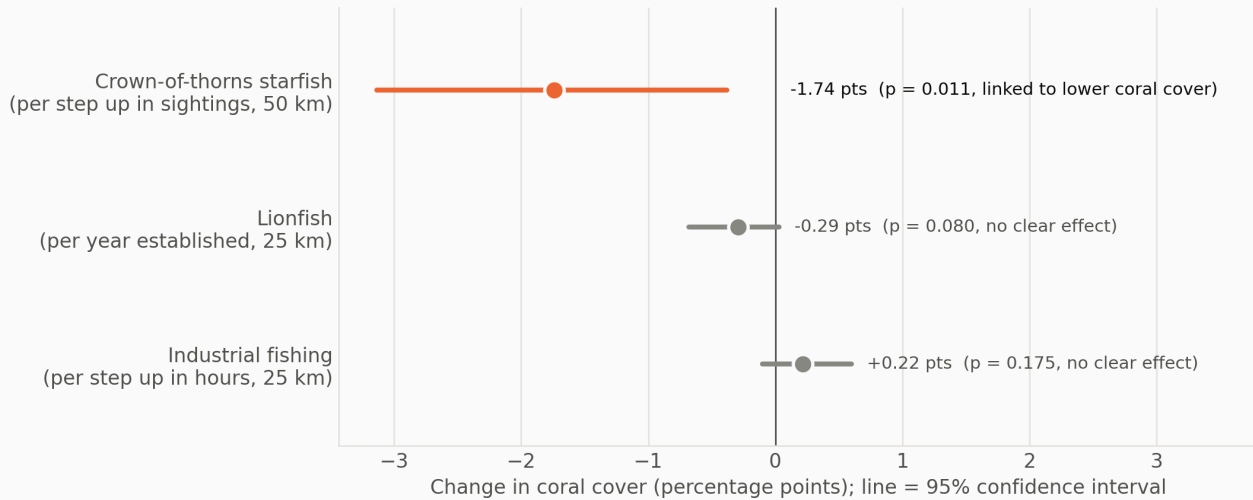
NOAA confirmed the global event ended in mid-2025, with no widespread bleaching in the 2025-26 southern summer ([GCRMN](#)). The model's 2026 figure (53% past 4 DHW, January to September) is consistent with that: 92% of those reefs are in the northern hemisphere, mostly Atlantic and Caribbean, with heat peaking in September 2026. That heat was still building when the data end, so 2026 is incomplete.

Invasive species and fishing

Of three non-heat threats, only crown-of-thorns starfish show a measurable link to lower coral cover. Reefs with more crown-of-thorns sightings within 50 km had 1.7 percentage points less coral cover per step up in sightings (95% range -3.1 to -0.4, $p = 0.011$, 3,745 surveys).

Which threats are linked to less coral?

Reefs compared within the same region and year, at the same heat stress

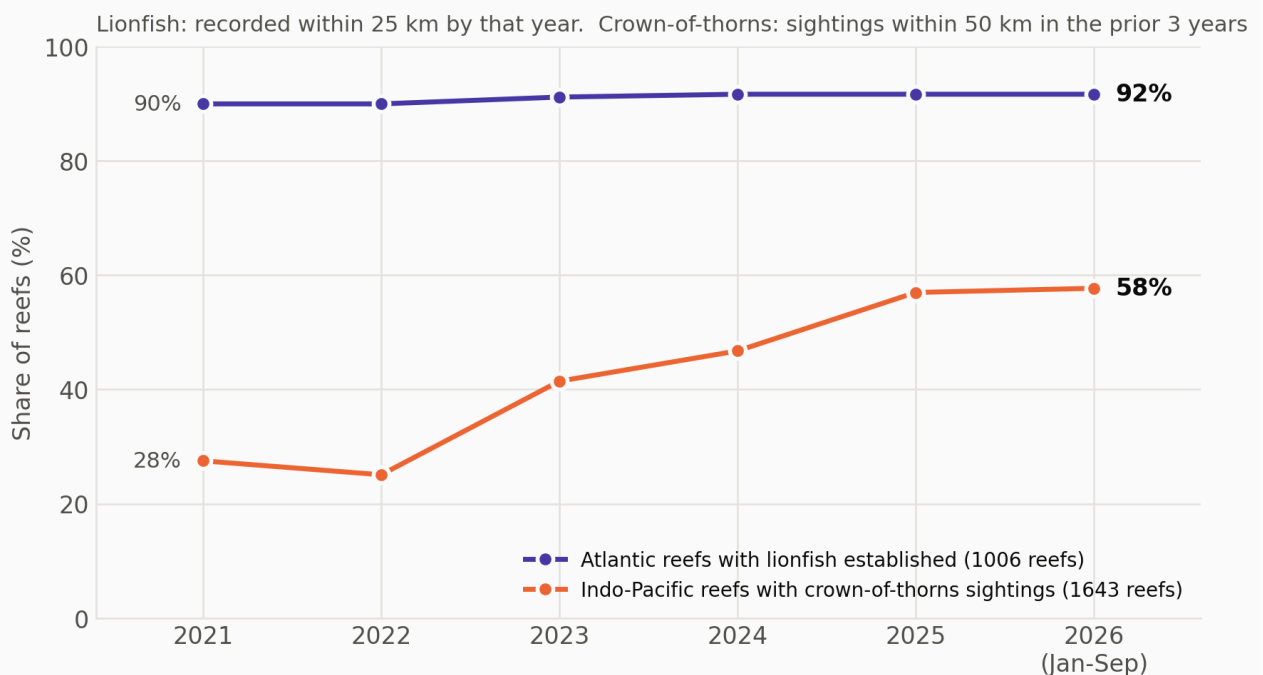


Heat stress is not in this chart: it drives bleaching (see risk curve). Sightings-based species data reflect where divers look as well as where species are.

- **Lionfish:** -0.3 points per year established within 25 km, but the range crosses zero ($p = 0.080$, 523 Atlantic surveys), so no effect is claimed.
- **Industrial fishing:** +0.2 points per step up in fishing hours within 25 km ($p = 0.175$, 4,268 surveys): no detectable link.

Each effect compares reefs in the same 5° region and year at the same heat stress, with uncertainty from resampling reefs 2,000 times. Only the crown-of-thorns effect is applied to the 2021-2026 threat file.

Invasive and outbreak species near reefs

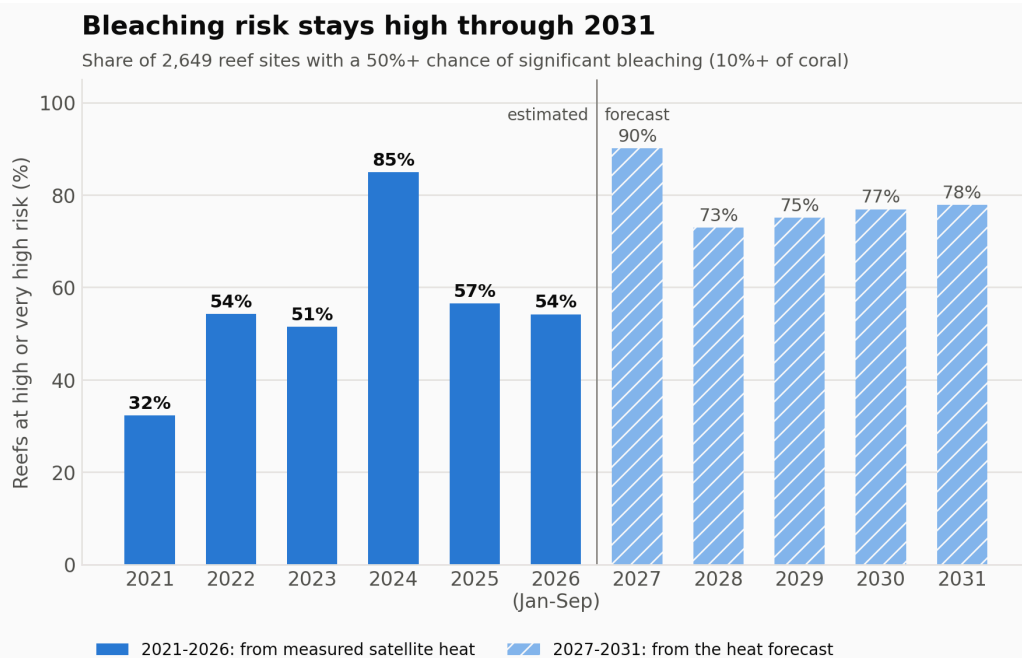


Data: GBIF occurrence records. A rise can reflect more divers reporting as well as more animals.

Crown-of-thorns sightings within 50 km rose from 28% of Indo-Pacific reefs in 2021 to 58% in 2026. Lionfish are established near 90-92% of Atlantic reefs throughout. Sightings reflect where divers look and report as well as where the animals are, so a rise partly reflects more reporting. Fishing data stop in 2020, so fishing columns for 2021-2026 are blank (no data), not zero.

Outlook, 2027-2031

Bleaching risk stays high through 2031: 90% of reef sites are at high or very high risk in 2027, then 73-78% each year to 2031. 2027 stands out because of a strong El Niño: NOAA's index for June-August 2026 is +1.80 ([NOAA CPC](#)).



2027 forecast is driven by the El Niño reading for Jun-Aug 2026 (ONI +1.8). Forecast years use one best-guess heat level per reef, so treat them as typical-year risk, not exact odds.

The outlook chains two models:

- Heat forecast.** Each reef's yearly peak heat back to 1985 was rebuilt from monthly sea temperature (checked against measured weekly heat: R^2 0.93, typically 0.8 DHW off). A ridge regression then forecasts peak heat 1-5 years ahead from the reef's recent peaks, its warming trend, its normal warmest-month temperature, latitude, ocean and El Niño.
- Bleaching model.** Each forecast peak goes through v2. The shape of each heat event is borrowed from the 25 most similar real 2021-2026 events in the same ocean. On real 2021-2026 events this shortcut ranked reefs almost the same as the full data (0.96 agreement) and ran about 3 points higher on average.

The heat forecast was tested by pretending to stand in each September from 2006 to 2024, forecasting ahead, and scoring against what happened:

Method (average of 1-5 years ahead)	Accuracy, passes 4 DHW (AUC)	Accuracy, passes 8 DHW (AUC)	Probability error at 8 DHW (Brier, lower is better)
Heat forecast (ridge, used)	0.756	0.737	0.113
XGBoost	0.724	0.709	0.118
Assume the last 5 years repeat (old scenario)	0.671	0.592	0.094

The forecast is clearly better at ranking which reefs will be hottest, but its probabilities for severe (8+ DHW) years are less accurate than the simple assumption. All methods predicted fewer hot years than occurred, because reefs keep getting hotter than their own past.

Limitations

The results rank reefs by risk well; the exact percentages should be read as approximate.

- **No surveys after 2020.** Predictions for 2021-2031 cannot yet be checked against divers' observations. Surveys from the 2023-2025 event would be the strongest test.
- **Percentages run high.** Even v2 predicted 1.4-1.9× more bleaching than divers found in later test years.
- **Forecast years are typical-year risk.** Each reef gets one best-guess heat level per year, so the forecast does not show the full range of possible years.
- **Heat beyond the training data.** 198 forecast reef-years are hotter than any surveyed reef the model learned from; these are flagged in the output.
- **Survey-based data has blind spots.** One 5 km satellite pixel stands in for each reef, heat is sampled weekly, and species sightings follow where divers go.
- **Associations, not proof.** The threat effects compare similar reefs; they do not prove cause and effect.

Output files

The map uses two files; the rest support them. All are in `data/processed/` of the project folder.

File	What it holds	Key columns
<code>reef_threats_2021_2026.csv</code>	Map layer: bleaching risk and threats per reef and year, 2021-2026	<code>bleaching_risk_band</code> , <code>p_significant_bleaching</code> , <code>predicted_pct_bleached</code> , <code>peak_dhw</code> , <code>threats_present</code>
<code>bleaching_forecast_2027_2031.csv</code>	Map layer: bleaching risk forecast per reef and year, 2027-2031	<code>risk_band</code> , <code>p_significant_bleaching</code> , <code>forecast_peak_dhw</code> , <code>p_dhw_over_4</code> , <code>p_dhw_over_8</code>
<code>bleaching_risk_v2_2021_2026.csv</code>	Bleaching predictions from the v2 model	<code>p_significant_bleaching</code> , <code>risk_band</code> , <code>assumed_survey_date</code>
<code>heat_forecast_2027_2031.csv</code>	Forecast peak heat per reef	<code>forecast_peak_dhw</code> , <code>p_dhw_over_4</code> , <code>p_dhw_over_8</code>
<code>heat_history_1985_2025.csv</code>	Yearly peak heat per reef, measured or estimated	<code>peak_dhw</code> , <code>peak_dhw_source</code>
<code>threat_effects.csv</code>	Effect of each threat on coral cover	<code>effect_pp</code> , <code>ci_low</code> , <code>ci_high</code> , <code>p_value</code>
<code>reef_stress_analysis.csv</code>	The 5,294 surveys with all joined data	one row per survey
<code>results_summary.csv</code>	Every number shown in the charts	<code>section</code> , <code>metric</code> , <code>value</code>

Risk bands in both map layers are `low` (under 25% chance of significant bleaching), `moderate` (25-50%), `high` (50-75%) and `very high` (75%+). Charts are in `charts/`, and outdated files from earlier stages are in `data/processed/old/`.

How to run the code

Everything runs from a terminal (for example the one in VS Code) with Python 3.10 or newer. Downloads are cached, so an interrupted step can simply be run again, and a step only needs rerunning when its inputs change.

1. Set up once

Open a terminal in the project folder and run (Windows shown; on Mac or Linux activate with `source .venv/bin/activate`):

```
python -m venv .venv
.venv\Scripts\activate
pip install -r requirements.txt
pip install xgboost scipy matplotlib
```

The first line makes a private Python environment for the project, the second switches it on (the prompt then starts with `(.venv)`), and the last two install the libraries. If Windows blocks the activate script, run `Set-ExecutionPolicy -Scope CurrentUser RemoteSigned` once and try again.

2. Get the data

- **Surveys:** download `global_bleaching_environmental.csv` from the BCO-DMO dataset page (see Sources) into `data/raw/`, keeping the file name.
- **Heat stress and sea temperature:** nothing to download by hand; the scripts fetch only the reef pixels they need from NOAA and cache them in `data/cache/`.
- **Fishing (optional):** download Global Fishing Watch monthly 0.1° fishing-effort files into `data/raw/gfw/`, or skip fishing with `python build_dataset.py --no-gfw`.

3. Run the steps in order

Step	Command	Makes
1	<code>python build_dataset.py</code>	<code>reef_stress_analysis.csv</code> : surveys joined with heat, fishing and species data
2	<code>python predict.py --years 2021 2022 2023 2024 2025</code> then <code>python predict.py --years 2026</code>	First-model predictions; also the list of reefs later steps use
3	<code>python timing_features.py</code>	<code>timing_features.csv</code> : heat timing around each survey
4	<code>python future_timing.py</code>	<code>future_timing_features.csv</code> : heat timing for 2021-2026
5	<code>python predict_v2.py --validate</code>	<code>bleaching_risk_v2_2021_2026.csv</code> , and prints the four test scores
6	<code>python threats.py</code>	<code>reef_threats_2021_2026.csv</code> and <code>threat_effects.csv</code>
7	<code>python fetch_monthly.py</code>	Monthly sea temperature since 1985, cached
8	<code>python forecast_heat.py</code>	<code>heat_history_1985_2025.csv</code> , <code>heat_forecast_2027_2031.csv</code> , <code>heat_forecast_validation.csv</code>
9	<code>python bleaching_forecast.py --check</code>	<code>bleaching_forecast_2027_2031.csv</code>
10	<code>python make_charts.py</code>	The seven charts in <code>charts/</code> and <code>results_summary.csv</code>

The slow steps are the downloads: step 1 fetches NOAA heat for about 1,100 reef pixels and can take several hours the first time, and step 7 takes about an hour. After that, every step reads from the cache; steps 3 and 4 take a few minutes and the rest under a minute each.

4. How the key code works

The snippets below are simplified from the project files, with comments for someone new to the code.

Heat before a survey (`fetch_noaa.py`). For each survey, the script takes the reef's satellite heat record and keeps the highest heat stress in the 30 days before the survey date:

```
recent = ts[(ts["time"] > end - pd.Timedelta(days=30)) & (ts["time"] <= end)]
max_dhw_30d = recent["CRW_DHW"].max() # peak heat stress in the 30 days
before the survey
```

The bleaching model (`predict_v2.py`). It first learns a severity score from the full % bleached, then turns that score into a chance of 10%+ bleaching:

```
class SeverityModel:
    def fit(self, X, pct):
        z = np.log((pct + 1) / (101 - pct)) # % bleached -> a
smooth severity score
        self.reg = make_pipeline(StandardScaler(), Ridge(alpha=1.0)).fit(X,
z)
        is_bleached = (pct >= 10).astype(int) # the yes/no target:
10%+ bleached
        self.cal = LogisticRegression().fit(self.reg.predict(X)[: , None],
is_bleached)
        return self

    def predict_proba(self, X): # chance of
significant bleaching
        return self.cal.predict_proba(self.reg.predict(X)[: , None])[: , 1]
```

`X` holds the features built by `design()`: heat in the last 30 days, the six heat-timing measures, the reef's ocean, ocean × heat, and distance from the equator.

Testing on unseen years (`predict_v2.py --validate`). Each year takes a turn as the test set while the model trains on the others:

```

for yr in sorted(d["year"].unique()):
    test = (d["year"] == yr).to_numpy()          # this year is
hidden from training
    model = SeverityModel().fit(X[~test], pct[~test]) # train on every
other year
    scores.append(roc_auc_score(y[test], model.predict_proba(X[test])))

```

The 2027-2031 outlook (`bleaching_forecast.py`). Forecast peak heat stands in for measured heat, and each forecast year borrows the event shape of similar real events:

```

fc["max_dhw_30d"] = fc["forecast_peak_dhw"]      # forecast heat
replaces measured heat
shape = shape_features(fc[["ocean", "max_dhw_30d"]], library) # timing from
the 25 most similar real events
X = design(pd.concat([fc, shape], axis=1))
fc["p_significant_bleaching"] = model.predict_proba(X)

```

5. Common problems

Message	Fix
<code>ModuleNotFoundError</code>	The environment is not active: run <code>.venv\Scripts\activate</code> , then <code>pip install</code> again if needed
<code>503 Server Error</code> from NOAA	The NOAA server is busy: run the same command again; finished downloads are kept
<code>timing_features.csv</code> is out of date	The survey table changed: rerun <code>python timing_features.py</code>
No <code>bleaching_risk_v2_*.csv</code> found	Run step 5 before steps 6 and 9

Sources

Data used by the project:

- [Global Coral Bleaching Database, van Woesik et al. \(BCO-DMO dataset 773466\)](#): diver surveys, 2013-2020
- [NOAA Coral Reef Watch](#) daily 5 km heat stress and monthly sea temperature, via [NOAA ERDDAP](#)

- [Global Fishing Watch](#) fishing effort, monthly 0.1°, 2012-2020
- [GBIF](#) occurrence records for *Pterois volitans*, *Pterois miles* and *Acanthaster*
- [NOAA CPC Oceanic Niño Index](#)

Pages cited in this report:

- [ICRI: 84% of the world's coral reefs impacted in the most intense global coral bleaching event ever](#)
- [GCRMN: The most intense global coral bleaching event on record has likely ended, NOAA confirms](#)